

Performance Optimization of Plate Fin Heat Sink Using ANSYS and Response Surface Methodology: A Review

Shaikh Yahya Gulam Rasool¹, A.V. Kodarkar²

¹Mtech Student, Shreeyash college of Engineering & Technology

²Professor, Shreeyash college of Engineering & Technology

Corresponding: Syahiya007@gmail.com

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Abstract

The increasing power density of modern electronic devices has intensified the need for efficient thermal management systems. Excessive heat generation adversely affects device performance, reliability, and lifespan, making effective cooling solutions essential. Among various thermal management techniques, plate fin heat sinks have emerged as one of the most widely adopted passive cooling devices due to their simple geometry, low manufacturing cost, and high heat dissipation capability. Over the years, significant research efforts have been devoted to enhancing the thermal performance of plate fin heat sinks through geometric optimization, material selection, and advanced computational analysis.

Computational Fluid Dynamics (CFD) and Finite Element Analysis (FEA) tools, particularly ANSYS Workbench and ANSYS Fluent, have become powerful platforms for investigating temperature distribution, airflow characteristics, thermal resistance, and heat transfer mechanisms within heat sink structures. Simultaneously, Response Surface Methodology (RSM) has gained considerable attention as a statistical optimization technique capable of developing predictive models and identifying optimal design parameters with reduced computational effort.

This review critically examines recent developments in the thermal performance optimization of plate fin heat sinks using ANSYS simulation and Response Surface Methodology. The influence of geometric parameters such as fin height, fin thickness, fin spacing, fin arrangement, and material properties on heat dissipation performance is discussed. Various numerical modeling approaches, optimization strategies, and validation techniques reported in the literature are analyzed. Furthermore, existing research gaps and future opportunities involving artificial intelligence, machine learning, topology optimization, and additive manufacturing are highlighted. The review demonstrates that the integration of ANSYS-based simulation and RSM provides an effective framework for achieving high-performance thermal management solutions for modern electronic applications.

1. Introduction

The continuous advancement of electronic technologies has led to significant increases in power density, processing speed, and miniaturization of electronic components. Modern electronic devices such as microprocessors, power modules, communication equipment, data centers, electric vehicle power electronics, LED lighting systems, and telecommunication devices generate substantial amounts of heat during operation. The effective removal of this heat has become one of the most critical challenges in electronic system design. Excessive operating temperatures can adversely affect system reliability, performance, efficiency, and lifespan, resulting in thermal failure, reduced operating stability, and increased maintenance costs. Consequently, the development of efficient thermal management systems has become an essential requirement for modern engineering applications [1,2].

Heat sinks are among the most widely employed passive cooling devices used to dissipate heat from electronic components to the surrounding environment. They function by increasing the available surface area for heat transfer, thereby enhancing the rate of heat dissipation through convection and radiation. Among the various heat sink configurations developed over the years, plate fin heat sinks have gained considerable attention due to their simple geometry, ease of manufacturing, low production cost, lightweight structure, and effective thermal performance. These characteristics make plate fin heat sinks suitable for a wide range of applications including consumer electronics, power electronics, aerospace systems, automotive electronics, and renewable energy devices [6,15].

The thermal performance of a plate fin heat sink is influenced by several geometric and operational parameters. Important design variables include fin height, fin thickness, fin spacing, fin length, fin profile, base plate thickness, material properties, airflow velocity, and heat load conditions. Appropriate selection and optimization of these parameters can significantly improve heat transfer performance while minimizing pressure losses and material consumption. However, due to the complex interaction among these parameters, designing an efficient heat sink remains a challenging engineering problem [10,15].

Traditionally, heat sink design relied heavily on experimental investigations and analytical approaches. Although experimental methods provide reliable results, they are often time-consuming, expensive, and limited in their ability to explore a large design space. Analytical models, on the other hand, generally involve simplifying assumptions that may reduce prediction accuracy for complex geometries and operating conditions. With the advancement of computational technologies, numerical simulation techniques have emerged as powerful alternatives for thermal analysis and optimization of heat sink systems [4,9].

Computational Fluid Dynamics (CFD) and Finite Element Analysis (FEA) have become widely adopted tools for investigating heat transfer and fluid flow phenomena in thermal management systems. Among the available simulation platforms, ANSYS Workbench and ANSYS Fluent are extensively used by researchers and engineers due to their ability to accurately model complex thermal-fluid interactions. ANSYS enables detailed evaluation of temperature distribution, velocity profiles, pressure drop characteristics, heat transfer coefficients, and thermal resistance under various operating conditions. Furthermore, simulation-based investigations significantly

reduce development time and cost by minimizing the need for extensive experimental testing [4,10,15].

In recent years, numerous studies have utilized ANSYS-based simulations to investigate the thermal and hydraulic performance of plate fin heat sinks. Abbas and Wang [1] explored displacement-based heat sink configurations and reported enhanced natural convection performance. Freegah et al. [4] and Hussain et al. [9] investigated the influence of fillet profiles and flow direction on heat transfer enhancement and demonstrated significant improvements in cooling effectiveness. Soloveva et al. [15] performed detailed numerical investigations of plate-fin heat sinks and highlighted the strong influence of geometric parameters on thermal-hydraulic performance. Similarly, Ismail et al. [10] conducted combined numerical and experimental investigations and confirmed the reliability of CFD-based thermal predictions.

The growing demand for improved thermal performance has encouraged researchers to explore innovative fin configurations and advanced optimization techniques. Studies involving variable-height fins, perforated fins, topology-optimized structures, and hybrid cooling configurations have demonstrated substantial improvements in heat dissipation capability [3,5,14,16]. Kamma et al. [11] emphasized the importance of considering both thermal performance and manufacturability during the optimization process, highlighting the practical challenges associated with advanced heat sink designs.

Despite the effectiveness of numerical simulation tools, the optimization of heat sink parameters remains a complex task due to the presence of multiple interacting variables. Evaluating every possible design combination through numerical simulation alone can be computationally expensive and time-intensive. To overcome these limitations, statistical optimization techniques have increasingly been integrated with simulation-based studies. Among these techniques, Response Surface Methodology (RSM) has emerged as one of the most widely used approaches for engineering optimization [2,12].

Response Surface Methodology is a collection of statistical and mathematical techniques used to develop predictive relationships between input variables and output responses. The methodology employs carefully designed experiments or simulations to generate data, which are then used to construct regression models representing the system behavior. In thermal management applications, RSM enables the identification of significant design parameters, prediction of thermal performance, and determination of optimal operating conditions with a reduced number of simulations or experiments. Techniques such as Central Composite Design (CCD) and Box–Behnken Design (BBD) are commonly employed to investigate the combined effects of geometric variables on heat sink performance [2,12].

Recent advancements have further expanded the scope of heat sink optimization through the integration of artificial intelligence, machine learning, and multi-objective optimization techniques. Nekahi et al. [13] combined numerical simulation, artificial neural networks, and optimization algorithms to enhance thermal performance of finned microchannel heat sinks. Wang et al. [16] utilized pseudo-three-dimensional topology optimization to improve thermal-hydraulic characteristics, while Yanagihara et al. [19] optimized fin angles using advanced flow resistance

models. These studies indicate a clear trend toward intelligent and data-driven thermal management solutions.

Although significant progress has been achieved in the design and optimization of plate fin heat sinks, several challenges remain unresolved. Many existing studies focus on individual geometric parameters rather than investigating their combined effects. Furthermore, limited research has explored the integration of ANSYS-based thermal simulations with Response Surface Methodology for comprehensive optimization of plate fin heat sink configurations. Additional research is also required to address practical considerations such as manufacturability, cost effectiveness, and multi-objective optimization involving thermal performance and pressure drop simultaneously.

Therefore, the primary objective of this review is to critically examine recent developments in the performance optimization of plate fin heat sinks using ANSYS simulation and Response Surface Methodology. The review focuses on numerical modeling approaches, effects of geometric parameters, optimization techniques, emerging research trends, and future opportunities. By consolidating recent findings from the literature, this work aims to provide researchers and engineers with a comprehensive understanding of current developments and potential directions for future investigation in plate fin heat sink optimization.

2. Plate Fin Heat Sink Fundamentals

Heat sinks are passive thermal management devices designed to dissipate heat generated by electronic components and transfer it to the surrounding environment. Their primary objective is to maintain the operating temperature of electronic systems within safe limits, thereby improving reliability, efficiency, and service life. Among the various heat sink configurations available, plate fin heat sinks have become one of the most widely used cooling solutions due to their simple design, ease of manufacturing, cost-effectiveness, and excellent thermal performance [6,15].

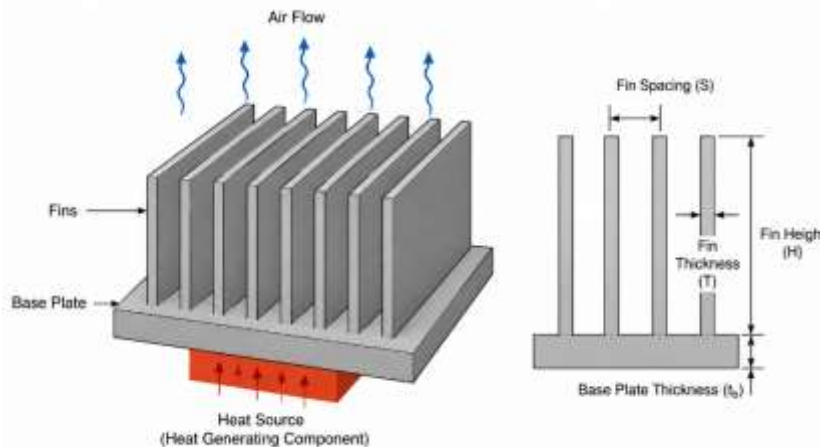


Figure 1: Schematic representation of a plate fin heat sink.

A plate fin heat sink typically consists of a base plate attached to a series of parallel fins. The base plate receives heat from the heat-generating component through conduction and distributes it uniformly across the fin structure. The fins increase the effective surface area available for heat transfer, thereby enhancing convective heat dissipation to the surrounding fluid, usually air. The

overall cooling performance of a heat sink depends on the combined effects of conduction within the solid structure and convection between the fin surfaces and the cooling medium [10].

The fundamental mechanism of heat transfer in a plate fin heat sink involves three sequential processes. Initially, heat generated by the electronic device is conducted through the base plate. Subsequently, heat is transferred from the base to the fins through conduction. Finally, the heat is dissipated from the fin surfaces to the surrounding environment by natural or forced convection. In some applications, radiation also contributes to heat transfer, although its effect is generally smaller compared to convection [1,6].

The thermal performance of a plate fin heat sink is governed by several geometric and operating parameters. Fin height is one of the most influential parameters because it directly affects the available surface area for heat transfer. Increasing fin height generally improves heat dissipation; however, excessive fin height may increase airflow resistance and reduce cooling effectiveness [17]. Similarly, fin thickness influences both thermal conduction and airflow characteristics. Thicker fins provide lower conduction resistance but reduce the available flow passage area, while thinner fins increase the number of fins that can be accommodated within a given space [2,12].

Fin spacing is another critical design variable that significantly affects airflow distribution and convective heat transfer. Small fin spacing increases heat transfer surface area but restricts airflow movement, whereas larger fin spacing improves airflow circulation at the expense of reduced heat transfer area. Therefore, an optimum fin spacing is required to achieve maximum cooling performance [15,17]. Additional parameters such as fin length, fin profile, fin arrangement, base thickness, material properties, and airflow velocity also play important roles in determining the overall thermal-hydraulic behavior of the heat sink [4,9].

Plate fin heat sinks can operate under either natural convection or forced convection conditions. In natural convection systems, airflow is generated by buoyancy forces arising from temperature differences between the heated surface and the surrounding fluid. Such systems are commonly used in low-power electronic devices due to their simplicity and low operating cost [1,6]. In forced convection applications, external fans or blowers are employed to increase airflow velocity, resulting in significantly higher heat transfer rates. Forced convection heat sinks are extensively used in high-power electronic systems, data centers, and industrial applications where enhanced cooling performance is required [10,12].

Material selection is another important consideration in heat sink design. Aluminum and copper are the most commonly used materials due to their high thermal conductivity and good manufacturability. Aluminum heat sinks are widely preferred because of their lightweight nature, corrosion resistance, and lower cost, whereas copper heat sinks provide superior thermal conductivity and enhanced heat spreading capabilities [15]. Recent studies have also explored advanced materials and composite structures to further improve thermal performance [20].

The thermal performance of a heat sink is commonly evaluated using parameters such as maximum temperature, thermal resistance, heat transfer coefficient, pressure drop, and fin efficiency. Maximum temperature is often considered the most important performance indicator because it directly reflects the effectiveness of heat dissipation. Thermal resistance represents the opposition

to heat flow and is commonly used to compare different heat sink configurations. Lower thermal resistance indicates better thermal performance and improved cooling capability [10,15].

Recent developments in plate fin heat sink technology have focused on enhancing thermal performance through innovative geometric modifications. Researchers have investigated variable-height fins, perforated fins, filleted fin profiles, topology-optimized structures, and hybrid cooling approaches to improve heat transfer characteristics [3,4,5,14,16]. These advanced designs aim to increase heat transfer area, improve airflow distribution, reduce pressure losses, and enhance overall cooling effectiveness.

The increasing complexity of heat sink geometries has also accelerated the adoption of numerical simulation techniques. Computational tools such as ANSYS Fluent and ANSYS Workbench enable detailed analysis of temperature distribution, airflow behavior, and thermal resistance under various operating conditions. These simulation capabilities provide valuable insights into heat transfer mechanisms and facilitate optimization of design parameters before prototype fabrication [4,10,15].

Understanding the fundamental principles governing plate fin heat sink operation is essential for developing efficient thermal management systems. The interaction between geometric parameters, material properties, and operating conditions determines the overall cooling performance. Consequently, systematic analysis and optimization of these factors remain key research areas in the development of high-performance heat sinks for modern electronic applications.

3. Recent Developments in Plate Fin Heat Sink Design

The increasing demand for efficient thermal management systems has motivated researchers to develop innovative plate fin heat sink designs capable of providing enhanced heat dissipation while maintaining acceptable pressure drop and manufacturing feasibility. Traditional straight plate fin heat sinks have been widely used for electronic cooling applications; however, their performance is often limited by airflow restrictions, thermal boundary layer development, and geometric constraints. Consequently, numerous investigations have focused on modifying fin geometry, fin arrangement, and flow characteristics to improve thermal performance [1,4,6].

One of the major research directions has been the modification of fin geometry to increase heat transfer efficiency. Haghighi et al. [6] proposed several new plate-fin heat sink configurations for natural convection cooling applications and demonstrated significant improvements in heat transfer performance compared with conventional straight-fin designs. Their study revealed that optimized fin arrangements can enhance airflow circulation and reduce thermal resistance, leading to improved cooling effectiveness.

Similarly, Abbas and Wang [1] investigated displacement-based plate fin heat sink configurations under natural convection conditions. The authors reported that introducing displacement between neighboring fins improved airflow movement and strengthened buoyancy-driven circulation. As a result, lower operating temperatures and improved heat dissipation rates were achieved compared to traditional heat sink configurations.

Another important development involves the introduction of fillet profiles at the junction between the fin and the base plate. Freegah et al. [4] performed CFD investigations on plate-fin heat sinks

incorporating various fillet geometries. Their results demonstrated that fillet profiles reduce flow separation and dead zones near the fin-base region, thereby enhancing heat transfer and reducing thermal resistance. Similar findings were reported by Hussain et al. [9], who examined the combined effects of flow direction and fillet geometry on heat sink performance. The study concluded that optimized fillet configurations significantly improve cooling effectiveness while maintaining acceptable pressure losses.

Recent studies have also explored variable-height fin arrangements as a means of improving airflow distribution and thermal performance. Gaikwad et al. [5] experimentally investigated plate fin heat sinks with variable fin heights under different orientation conditions. The results indicated that variable-height configurations promote more uniform airflow distribution and improve natural convection heat transfer. The authors observed substantial reductions in maximum temperature compared to conventional uniform-height fin designs.

The incorporation of perforations, bulges, and surface modifications has emerged as another promising strategy for thermal enhancement. Chowdhury et al. [3] investigated plate pin-fin heat sinks with different pin shapes combined with perforations and bulged surfaces. Their findings revealed that these modifications enhance fluid mixing and disrupt thermal boundary layers, resulting in improved heat transfer rates. Such geometric modifications have gained considerable attention because they improve cooling performance without significantly increasing system size. Advancements in topology optimization have further expanded the possibilities for heat sink design. Wang et al. [16] applied pseudo-three-dimensional topology optimization techniques to improve the thermal-hydraulic performance of plate-fin heat exchangers. Their optimized fin structures exhibited superior heat transfer capability and reduced flow resistance compared with conventional designs. Similarly, Pandey and Lee [14] developed topology-optimized liquid-cooled heat sinks and demonstrated substantial improvements in thermal efficiency through intelligent material distribution strategies.

Manufacturability has become an increasingly important consideration in modern heat sink design. While advanced geometric configurations often provide enhanced thermal performance, they may also introduce manufacturing challenges and increased production costs. Kamra et al. [11] investigated plate-fin heat sink configurations considering both thermal performance and manufacturability constraints. Their optimization approach successfully identified designs that achieved improved cooling performance while remaining practical for industrial manufacturing processes.

In addition to air-cooled heat sinks, researchers have explored heat sink designs integrated with thermoelectric cooling systems and liquid cooling technologies. He et al. [7] optimized plate-fin heat exchanger structures for thermoelectric cooling applications and reported improved heat dissipation performance through geometric modifications. Yanagihara et al. [19] investigated fin angle optimization in water-cooled plate-fin heat sinks and demonstrated significant thermal improvements through adjustment of fin orientation.

The influence of fin structural parameters continues to be an active area of research. Xie et al. [17] numerically investigated the effects of fin height, fin thickness, and fin spacing on heat dissipation

performance. Their study confirmed that these geometric parameters strongly influence maximum temperature, thermal resistance, and overall cooling effectiveness. The authors emphasized the importance of parameter optimization for achieving high-performance thermal management systems.

Recent developments indicate a clear transition from conventional trial-and-error design approaches toward simulation-driven and optimization-based methodologies. Modern heat sink research increasingly combines Computational Fluid Dynamics (CFD), Finite Element Analysis (FEA), topology optimization, artificial intelligence, and statistical optimization techniques to develop highly efficient cooling solutions. These approaches not only improve thermal performance but also reduce development time and cost, making them attractive for industrial applications [13,14,16].

Overall, recent advancements in plate fin heat sink design have demonstrated that substantial improvements in thermal performance can be achieved through geometric modifications, innovative fin configurations, and advanced optimization techniques. The integration of computational modeling and optimization tools has enabled the development of next-generation heat sink designs capable of meeting the increasing thermal demands of modern electronic systems.

Table 1. Summary of recent studies on plate fin heat sink optimization

Ref.	Author & Year	Method	Key Parameter Investigated	Major Findings
[1]	Abbas and Wang (2020)	Numerical	Fin displacement	Improved natural convection performance
[4]	Freegah et al. (2020)	CFD	Fillet profile	Reduced thermal resistance
[5]	Gaikwad et al. (2025)	Experimental	Variable fin height	Enhanced heat transfer
[10]	Ismail et al. (2024)	Experimental + CFD	Plate-fin design	Validated numerical predictions
[11]	Kamma et al. (2025)	Optimization	Fin configuration	Improved thermal performance and manufacturability
[15]	Soloveva et al. (2024)	Numerical	Fin geometry	Improved thermal-hydraulic behavior
[17]	Xie et al. (2026)	Numerical	Fin height, thickness, spacing	

4. ANSYS-Based Numerical Analysis of Plate Fin Heat Sinks

The rapid advancement of computational technologies has significantly transformed the design and optimization of thermal management systems. Numerical simulation techniques have become indispensable tools for analyzing heat transfer and fluid flow phenomena in plate fin heat sinks.

Compared with traditional experimental investigations, numerical methods offer reduced development cost, shorter analysis time, greater flexibility in design modification, and the ability to investigate complex geometries under a wide range of operating conditions. Among the available computational platforms, ANSYS Workbench and ANSYS Fluent are the most widely used software packages for thermal and fluid flow analysis of heat sink systems [4,10,15].

ANSYS-based numerical analysis enables researchers to evaluate temperature distribution, airflow behavior, pressure drop characteristics, thermal resistance, and heat transfer coefficients within heat sink structures. The simulation process generally begins with the development of a three-dimensional geometric model of the heat sink, followed by assignment of material properties, mesh generation, application of boundary conditions, numerical solution, and post-processing of results. These capabilities allow detailed investigation of thermal performance without the need for extensive prototype fabrication and testing [10].

The geometry of a plate fin heat sink is typically created using ANSYS DesignModeler, SpaceClaim, or external CAD software before being imported into the simulation environment. Important geometric parameters include fin height, fin thickness, fin spacing, base thickness, and fin number. Once the geometry is developed, material properties such as thermal conductivity, density, and specific heat are assigned. Aluminum and copper are the most commonly used materials due to their favorable thermal characteristics [15].

Mesh generation is one of the most critical steps in numerical simulation because it directly influences solution accuracy and computational efficiency. Fine mesh structures are generally employed near fin surfaces and heat transfer regions where large temperature gradients are expected. Grid independence studies are often conducted to ensure that simulation results are independent of mesh size and numerical discretization errors. A properly generated mesh provides reliable predictions while minimizing computational cost [4,9].

Boundary conditions play a vital role in determining simulation accuracy. Typical thermal boundary conditions include constant heat flux, constant temperature, or volumetric heat generation applied at the base of the heat sink. Fluid flow boundary conditions may include specified inlet velocity, pressure outlet, ambient temperature, and turbulence parameters. Depending on the application, simulations may be performed under natural convection or forced convection conditions. Accurate representation of operating conditions is essential for obtaining realistic thermal predictions [10,15].

Numerous researchers have successfully employed ANSYS simulations to investigate the thermal performance of plate fin heat sinks. Freegah et al. [4] utilized CFD analysis to evaluate heat transfer enhancement in plate-fin heat sinks with fillet profiles. Their results demonstrated that fillet geometries improve airflow characteristics and reduce thermal resistance. The study highlighted the ability of ANSYS Fluent to accurately capture complex flow structures and temperature distributions within heat sink channels.

Similarly, Hussain et al. [9] performed numerical investigations to analyze the effects of flow direction and fillet profile on heat transfer enhancement. Their simulations revealed significant improvements in cooling performance resulting from optimized fin configurations and flow

arrangements. The study demonstrated the importance of CFD-based approaches in identifying efficient thermal management solutions.

Soloveva et al. [15] conducted detailed numerical investigations on the thermal and hydraulic characteristics of plate-fin heat sinks. Their analysis focused on the influence of geometric parameters and operating conditions on heat transfer performance and pressure drop. The authors reported that numerical simulation provides valuable insights into thermal-hydraulic behavior and can effectively guide the design optimization process.

Ismail et al. [10] combined experimental and numerical approaches to investigate the hydrothermal performance of plate-fin heat sinks. The comparison between simulated and experimental results showed good agreement, confirming the reliability and accuracy of ANSYS-based thermal analyses. Such validation studies are important because they establish confidence in simulation results and support their use in engineering design applications.

Recent developments in numerical analysis have expanded beyond conventional plate fin configurations. He et al. [7] investigated optimized plate-fin heat exchanger designs for thermoelectric cooling systems using numerical simulation techniques. Their findings demonstrated that geometric optimization can substantially improve heat dissipation performance and system efficiency. Similarly, Wang et al. [16] employed topology optimization integrated with numerical analysis to enhance thermal-hydraulic performance of plate-fin heat exchangers.

Advanced numerical studies have also incorporated artificial intelligence and machine learning approaches. Nekahi et al. [13] integrated Computational Fluid Dynamics simulations with artificial neural networks and multi-objective optimization techniques for heat sink design improvement. Their work demonstrated the potential of combining simulation data with intelligent optimization algorithms to achieve superior thermal performance.

One of the primary advantages of ANSYS-based thermal analysis is the ability to visualize temperature contours, velocity vectors, pressure distributions, and heat flux patterns. These graphical outputs provide detailed understanding of heat transfer mechanisms and enable identification of thermal hotspots, airflow recirculation zones, and regions of poor cooling performance. Such information is often difficult or impossible to obtain through experimental methods alone [4,15].

Despite its advantages, numerical simulation also has certain limitations. The accuracy of simulation results depends on the quality of geometric modeling, mesh generation, material property specification, turbulence modeling, and boundary condition selection. Inadequate representation of these factors may lead to discrepancies between simulated and experimental results. Therefore, validation against experimental data remains an essential component of simulation-based investigations [10].

Overall, ANSYS-based numerical analysis has become a powerful and reliable tool for evaluating the thermal performance of plate fin heat sinks. The ability to accurately predict temperature distribution, airflow behavior, thermal resistance, and heat transfer characteristics makes ANSYS an indispensable platform for thermal management research. Furthermore, numerical simulation provides the foundation for advanced optimization techniques such as Response Surface

Methodology, enabling systematic investigation of design parameters and development of high-performance heat sink configurations.

5. Influence of Geometric Parameters on Thermal Performance

The thermal performance of a plate fin heat sink is strongly influenced by its geometric configuration. Parameters such as fin height, fin thickness, fin spacing, fin profile, and fin arrangement directly affect heat transfer area, airflow distribution, thermal resistance, and pressure drop. Optimization of these geometric variables is essential for achieving maximum heat dissipation while maintaining acceptable hydraulic performance. Numerous experimental and numerical investigations have demonstrated that even small modifications in heat sink geometry can produce significant improvements in cooling effectiveness [2,12,15,17].

5.1 Effect of Fin Height

Fin height is one of the most critical design parameters because it determines the available surface area for convective heat transfer. Increasing fin height generally enhances heat dissipation by increasing the heat transfer area exposed to the cooling medium. Higher fins provide greater contact between the fin surface and surrounding air, resulting in improved cooling performance and lower operating temperatures [5,17].

However, the relationship between fin height and thermal performance is not always linear. Excessive fin height may increase airflow resistance, reduce air circulation within fin channels, and increase manufacturing complexity. Xie et al. [17] reported that although increasing fin height initially reduces maximum temperature, further increases beyond an optimum value provide diminishing thermal benefits. Similarly, Gaikwad et al. [5] demonstrated that variable-height fin configurations can improve airflow distribution and enhance natural convection heat transfer compared with conventional uniform-height designs.

Therefore, determining the optimum fin height is essential for balancing heat transfer enhancement and airflow effectiveness.

5.2 Effect of Fin Thickness

Fin thickness influences both heat conduction within the fin structure and airflow characteristics between neighboring fins. Thicker fins offer lower conduction resistance and improve heat spreading from the base plate to the fin tips. However, increasing fin thickness reduces the available flow passage area and may restrict airflow movement, especially under natural convection conditions [2,12].

Ahmed [2] investigated the thermal optimization of ribbed flat-plate fin heat sinks and reported that moderate fin thickness provides the best compromise between thermal conduction and convective heat transfer. Khattak and Muhammad [12] also observed that excessive fin thickness can increase pressure losses while providing only marginal improvements in heat dissipation.

Consequently, an optimum fin thickness must be selected to maximize overall thermal performance while minimizing airflow obstruction.

5.3 Effect of Fin Spacing

Fin spacing governs airflow distribution and thermal boundary layer development within the heat sink channels. It is considered one of the most influential parameters affecting convective heat

transfer performance. Small fin spacing increases the total heat transfer surface area but restricts airflow movement and may lead to thermal boundary layer interference. Conversely, larger fin spacing improves airflow circulation but reduces the total available surface area for heat transfer [15,17].

Numerical investigations conducted by Soloveva et al. [15] and Xie et al. [17] demonstrated that there exists an optimum fin spacing that maximizes cooling performance. Their studies revealed that inappropriate fin spacing can significantly increase maximum temperature and thermal resistance. Therefore, optimization of fin spacing is crucial for achieving efficient heat dissipation.

5.4 Effect of Fin Profile

The shape and profile of fins significantly influence thermal-hydraulic performance. Traditional straight fins are commonly used because of their simple geometry and ease of manufacturing. However, recent studies have explored alternative fin profiles including filleted fins, tapered fins, perforated fins, bulged fins, and topology-optimized structures to improve heat transfer characteristics [3,4,9].

Freegah et al. [4] investigated plate-fin heat sinks with fillet profiles and reported enhanced heat transfer due to smoother airflow transitions and reduced flow separation. Hussain et al. [9] similarly demonstrated that fillet geometries improve thermal performance by reducing stagnant flow regions near the fin-base junction.

Chowdhury et al. [3] examined perforated and bulged fin configurations and observed substantial improvements in heat transfer resulting from enhanced fluid mixing and disruption of thermal boundary layers. These studies indicate that innovative fin profiles can significantly improve cooling effectiveness without major increases in heat sink size.

Effect of Fin Arrangement and Orientation

The arrangement and orientation of fins influence airflow pathways and heat transfer efficiency. Different fin arrangements can alter fluid circulation patterns, temperature distribution, and pressure losses within the heat sink. Researchers have investigated staggered, aligned, displaced, and variable-height fin arrangements to enhance cooling performance [1,5].

Abbas and Wang [1] demonstrated that displacement-based fin arrangements improve natural convection performance by promoting more effective airflow circulation. Similarly, Gaikwad et al. [5] reported that heat sink orientation significantly affects natural convection cooling, with certain orientations providing superior heat transfer performance.

Optimization of fin arrangement and orientation is therefore important for maximizing thermal performance under both natural and forced convection conditions.

5.6 Effect of Fin Angle and Topology Optimization

Recent advancements in heat sink design have focused on topology optimization and fin angle modification. Yanagihara et al. [19] investigated fin angle optimization in water-cooled plate-fin heat sinks and demonstrated significant improvements in heat transfer performance through strategic adjustment of fin orientation. The optimized fin angles improved coolant distribution and reduced thermal resistance.

Similarly, Wang et al. [16] applied pseudo-three-dimensional topology optimization techniques to redesign plate-fin structures and reported substantial enhancement in thermal-hydraulic performance. Pandey and Lee [14] further demonstrated that topology-optimized fin configurations can significantly improve cooling efficiency while reducing material usage. These studies highlight the growing importance of advanced geometric optimization techniques in next-generation heat sink development.

5.7 Comparative Influence of Geometric Parameters

Based on the reviewed literature, fin spacing and fin height are generally identified as the most influential parameters affecting maximum temperature and thermal resistance [15,17]. Fin thickness primarily affects conduction characteristics, while fin profile and fin arrangement influence airflow behavior and heat transfer enhancement [1,3,4]. The combined optimization of these parameters is essential for achieving superior thermal performance.

Table 2. Effect of geometric parameters on thermal performance

Parameter	Effect on Tmax	Effect on Thermal Resistance	Effect on Airflow
Fin Height	Decreases Tmax up to optimum value	Decreases	May increase resistance
Fin Thickness	Moderate decrease	Decreases conduction resistance	Reduces flow area
Fin Spacing	Strong influence	Strong influence	Controls airflow distribution
Fin Profile	Improves heat transfer	Reduces	Improves mixing
Fin Arrangement	Improves cooling	Reduces	Enhances circulation
Fin Angle	Improves heat dissipation	Reduces	Improves flow path

Because these geometric variables interact with one another, independent optimization of individual parameters may not yield the best overall design. Consequently, statistical optimization techniques such as Response Surface Methodology (RSM) are increasingly employed to investigate the combined effects of multiple design variables and identify optimal parameter combinations [2,12].

The collective findings from recent studies indicate that careful geometric optimization can significantly improve heat dissipation capability, reduce thermal resistance, and enhance the overall performance of plate fin heat sinks used in modern thermal management applications.

6. Research Gaps and Future Research Directions

The extensive body of literature on plate fin heat sinks demonstrates significant progress in thermal performance enhancement through geometric modifications, numerical simulations, and optimization techniques. Researchers have investigated various fin configurations, cooling strategies, and computational approaches to improve heat dissipation efficiency. Despite these

advancements, several research challenges and knowledge gaps remain unresolved, providing opportunities for further investigation and development.

One of the major limitations observed in the existing literature is the lack of studies integrating numerical simulation and statistical optimization within a comprehensive framework. While numerous researchers have independently employed ANSYS-based thermal analysis [4,10,15] and optimization techniques [2,12], relatively few studies have systematically combined these approaches to simultaneously evaluate multiple geometric parameters and their interactions. Such integration is essential for achieving more accurate and efficient design optimization.

Another significant research gap concerns the optimization objectives considered in previous studies. Most investigations focus primarily on minimizing maximum temperature or thermal resistance. However, practical heat sink design requires simultaneous consideration of multiple performance criteria, including pressure drop, airflow resistance, manufacturing cost, material consumption, weight, and reliability. The development of multi-objective optimization frameworks capable of balancing these competing requirements remains limited in current research [11,13].

Although several studies have examined the influence of individual geometric parameters such as fin height, fin thickness, and fin spacing [15,17], fewer investigations have explored their combined effects under realistic operating conditions. Interactions among design variables can significantly influence thermal performance, and neglecting these interactions may result in suboptimal designs. Therefore, comprehensive parametric investigations involving multiple geometric variables are required to better understand their collective impact on heat transfer behavior.

The majority of available research has concentrated on conventional straight-fin and modified plate-fin configurations operating under either natural or forced convection conditions. Comparatively fewer studies have investigated hybrid cooling systems, advanced fin structures, and next-generation thermal management solutions. Emerging technologies such as topology-optimized fins, biomimetic heat sink structures, perforated fins, and variable-geometry configurations offer promising opportunities for further thermal enhancement and require additional investigation [3,14,16,19].

Another important limitation is the insufficient experimental validation reported in many numerical studies. Although ANSYS and CFD simulations provide detailed insights into thermal-fluid behavior, simulation results are inherently dependent on assumptions related to boundary conditions, turbulence models, material properties, and mesh quality. Experimental validation remains essential to establish confidence in numerical predictions and ensure practical applicability of optimized designs [10]. Future research should emphasize combined experimental and numerical approaches to improve result reliability.

The application of artificial intelligence and machine learning in heat sink optimization remains relatively limited despite its growing importance in engineering design. Recent work by Nekahi et al. [13] demonstrated the potential of integrating artificial neural networks with numerical simulations and optimization algorithms. However, the use of machine learning models for thermal

prediction, real-time optimization, and intelligent heat sink design is still in its early stages. Further research is needed to explore advanced data-driven approaches capable of handling complex nonlinear thermal systems.

Manufacturing considerations represent another area requiring greater attention. Many optimized heat sink configurations reported in the literature involve complex geometries that may be difficult or expensive to manufacture using conventional fabrication techniques. Although recent advances in additive manufacturing have expanded design possibilities, limited research has examined the relationship between thermal performance optimization and manufacturing feasibility. Future studies should focus on developing designs that achieve both superior thermal performance and practical manufacturability [11].

Material selection is also an important area for future development. Most current plate fin heat sinks are manufactured using aluminum or copper because of their high thermal conductivity and favorable mechanical properties. However, emerging materials such as metal matrix composites, graphene-enhanced materials, and advanced alloys may offer improved thermal performance while reducing weight and manufacturing costs. Further investigations are required to evaluate the suitability of these materials for next-generation thermal management systems.

The increasing demand for compact and high-power electronic devices has also highlighted the need for advanced cooling technologies capable of managing extremely high heat fluxes. Future research should investigate the integration of plate fin heat sinks with phase change materials, thermoelectric cooling systems, liquid cooling technologies, and hybrid thermal management approaches. Such systems have the potential to provide substantially higher cooling performance compared with conventional air-cooled heat sinks [7,20].

Based on the identified research gaps, future studies should focus on developing integrated optimization frameworks combining ANSYS simulation, Response Surface Methodology, artificial intelligence, and multi-objective optimization techniques. These approaches can enable systematic exploration of complex design spaces and facilitate the development of high-performance thermal management solutions. Additionally, increased emphasis should be placed on experimental validation, advanced materials, topology optimization, additive manufacturing, and hybrid cooling technologies.

In summary, although substantial progress has been achieved in plate fin heat sink design and optimization, significant opportunities remain for further advancement. Addressing the identified research gaps will contribute to the development of more efficient, reliable, and cost-effective thermal management systems capable of meeting the growing cooling demands of modern electronic applications.

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